



Article

Digital Transformation and the Quality of Accounting Information Systems in the Public Sector: Evidence from Developing Countries

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Abstract: This study explores how digital transformation affects the accounting and auditing profession and the quality of the accounting information system in governmental units in two developing contexts in the Middle East and North African (MENA) region—Egypt and Saudi Arabia. We collected data by electronically surveying governmental units' employees in the two countries and analyzed them using linear regression analyses. Interestingly, we found a negative effect of digital transformation on the quality of accounting information systems, which could be interpreted concerning the premature status of digital technologies in emerging markets. However, the negative effect of digital transformation was lower in Egypt compared to Saudi Arabia. This study contributes to the literature by focusing on the governmental sector and investigating a new context—the MENA region—which has rarely been examined in previous studies. Further, it has implications for regulators in the public sector for developing governmental financial reports by supporting the digital transformation process.

Keywords: digital transformation; accounting; auditing; public sector; Egypt; Saudi Arabia



Academic Editors: Hassan Obeid and Zied Ftiti

Received: 12 January 2025

Revised: 31 January 2025

Accepted: 25 February 2025

Published: 28 February 2025

Citation: Hamdy, A., Diab, A., & Eissa, A. M. (2025). Digital Transformation and the Quality of Accounting Information Systems in the Public Sector: Evidence from Developing Countries. *International Journal of Financial Studies*, 13(1), 30. <https://doi.org/10.3390/ijfs13010030>

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1. Introduction

Recent crises such as the COVID-19 pandemic have increased reliance on information and communication technology to reduce their negative effects, which has expanded the scope of technology application in the daily lives of individuals and organizations (Rashid et al., 2021; Jabor & Hamdan, 2023). The concept of digitalization has recently emerged as an infrastructure to change business models and create revenue-generating opportunities (Alenezi & Akour, 2023; Begkos et al., 2024; Alanazi & Alenezi, 2024). Relatedly, the concept of digital transformation is being used for a strategic reorientation of business models (F. Fang et al., 2024) or achieving a comprehensive integration of digital tools to restructure business processes and services, which is vital to enhance relationships with stakeholders (Cavicchi & Vagnoni, 2023).

In the government sector, digital transformation represents a radical change in providing government services by fully utilizing the Internet and digital technology, which is crucial to improve the performance of governmental agencies, increase the quality of administration and public services (Aminah & Saksono, 2021), and enhance the quality of life for citizens through more responsive and transparent management and governance (Ciancarini et al., 2024). This is considered important to shorten or facilitate administrative

procedures, reduce potential errors, and improve the accuracy of governmental information and reports (Abhishek et al., 2024b). This change is aided by digital transformation technologies, such as enterprise resource planning systems, blockchain, Internet of Things, robotic process automation, and artificial intelligence (Lang, 2021; Jabor & Hamdan, 2023).

Accounting and auditing professions are highly affected by digital transformation. Digital accounting systems indicate the use of digital technology and software to manage financial data, transactions, and various reports (Al-Hattami et al., 2024). This has radically transformed the way accounting and auditing functions were conventionally conducted (Yigitbasioglu et al., 2023; Nguyen et al., 2024; Pedroso & Gomes, 2024).

However, the impact of digital transformation on accounting information systems in the public sector still needs more investigation, as current research focuses on the benefits and challenges of adopting digital transformation in the accounting and auditing professions (Fernandez & Aman, 2018; Kaya et al., 2019; Otia & Bracci, 2022; Potryvaieva et al., 2022; Aboelfotoh et al., 2024; F. Fang et al., 2024; Pedroso & Gomes, 2024). Moreover, the existing research on the governmental sector has addressed the administrative and institutional reforms necessary to reduce digital transformation risks, highlighting the institutional factors that enhance the success of digital transformation in governmental agencies (Eom & Lee, 2022; Kenetey & Popesko, 2024; Dos Santos et al., 2024). There is a scarcity of research in the field of digital transformation and the quality of accounting information systems in governmental units in emerging markets, especially Egypt and Saudi Arabia. The unique institutional context of the two countries makes them worthy of special investigation. We noted that the excessive taxation in Egypt is threatening the governmental plan for digital transformation because it discourages e-payments, de-incentivizes new market entrants, and precludes the formalization of the prevalent informal market.¹ In Saudi Arabia, the existing challenges for digital transformation include credit card usage concerns and customers' fears regarding the security of electronic channels (Gull et al., 2022). Thus, our study examines how digital transformation affects accounting and auditing professions and the quality of accounting information systems in governmental units by bringing evidence from two emerging markets in the MENA region.

Therefore, the current study derives its importance from filling an existing research gap in the field of digital transformation and the quality of accounting information systems in developing countries' governmental units. The researchers conducted a survey of governmental employees' perceptions on the impact of digital transformation on the quality of accounting information systems in governmental units in Egypt and Saudi Arabia. The findings indicate that digital transformation had a negative effect on the quality of accounting information systems when this effect was examined for the whole sample (Egypt and KSA). This finding implies that during the early stages of using digital technologies in emerging markets, the digital transformation level is relatively low, and the existing complexities arising from the newly used digital systems may negatively affect the current accounting information systems (F. Fang et al., 2024). Moreover, our analysis revealed that the negative effect of digital transformation was lower in Egypt, indicating that Egypt had personnel more qualified in using and benefiting from digital technology compared to Saudi Arabia. These findings contribute to the literature by focusing on the governmental sector and investigating a new context—the MENA region—rarely examined in previous studies. Moreover, our findings can have implications for regulators in the public sector who should develop governmental financial reports utilizing the digital transformation process. They should manage the challenges brought about by digital transformation in accounting and auditing professions.

The remainder of the study is organized as follows. Section 2 provides a contextual background of the study. Section 3 reviews the literature and develops the study hypothesis.

Section 4 summarizes the research methods. Section 5 displays and discusses the research results. Finally, Sections 6 and 7 discuss the findings and conclude the paper, respectively.

2. A Background

In Egypt, developing a digital infrastructure is one of the foundations on which the governmental vision of Digital Egypt 2030 is based. Following Prime Ministerial Decree No. 2374 for 2017, all units of the state administrative apparatus, local administration units, and public service agencies have implemented the Government Financial Management Information System (GFMIS). This system is an electronic system that links all public agencies and aims to raise the level of financial transparency in the public sector units. The government's Communications and Information Technology Strategy also supports the goals of Egypt's Vision 2030 by developing the infrastructure for information and communication technology, enhancing digital inclusion, achieving financial inclusion, encouraging innovation, combating corruption, and ensuring information security.²

Moreover, it was crucial to issue legislative and legal amendments regarding financial transactions, intellectual property rights, trademarks, electronic business, electronic signatures, hacking, and electronic crimes in the Egyptian business environment. The Arab Republic of Egypt has taken some steps in this regard, including developing the information technology and communications sector, keeping pace with technological developments, and issuing some laws to regulate this sector, such as the Electronic Crimes Law, Intellectual Property Laws, Consumer Protection Law, Electronic Signature Law, and Personal Data Protection Law. They also issued the executive regulations of the Law on Combating Information Technology Crimes.³

In this way, financial regulations for budget and accounts were issued in 2021. Articles No. 164–173 regulated using and implementing the government financial management information system. Article No. 502 stated that “accounting operations shall be recorded in books and records, both in paper and electronic forms, along with the instructions of the Ministry of Finance”. Article No. 50 of the Unified General Budget Law stipulated that “the administrative body shall maintain accounting documents, and books in paper and electronic forms”.

Vision 2030 of Saudi Arabia indicates that the country wants to be one of the top digital governments in the world by 2030. For that purpose, the digital government policy sought to support the advancement of governmental units' strategic plans in line with the central digital government strategy. One of the notable applications of the digital government in Saudi Arabia is the wide launch and use of *Absher*, which is a smartphone application and web portal permitting local citizens and foreign residents to virtually approach various governmental services smoothly. Considering the importance of governance for the success of digital transformation, the Digital Government Authority in the Kingdom of Saudi Arabia relies on the standards for digital transformation to measure public entities' commitment to the requirements of digital transformation. In 2022, the digital transformation standards emphasized the necessity for implementing the main systems of government resource planning and the existence of a guide for these systems that is circulated to users after training them on these systems.⁴

3. Literature Review and Hypothesis Development

This work is informed by a set of theories such as the technology acceptance model (TAM), contingency theory, and the agency theory. TAM is a theoretical structure that aims to provide insights into the various aspects that influence individuals' adoption of technology, as well as the underlying motivations for their acceptance (Gündoğan & Keçeci, 2024). It was proposed by Davis (1989), and it focuses on two primary factors

influencing technology adoption: perceived usefulness and ease of use (Limna et al., 2023). TAM posits that if users believe a technology is useful and easy to use, they are more likely to adopt and use it. This model has been widely used to predict user acceptance of various technologies and understand user behavior toward technology. According to Davis (1989), one of the elements that contribute to the adoption of information technology is what is known as “perceived usefulness”. Another important factor that influences an information system’s acceptability is its ease of use. People with a positive attitude toward technology have a greater chance of accepting the system and reaping the benefits of using it (Kocsis et al., 2022).

From the contingency theory perspective, a proper implementation of any strategy depends largely on certain factors such as employee behavior (Woodward, 1958; Nicolás-Agustín et al., 2022). Thus, human resources must be well managed consistent with the digital transformation strategy for such a strategic implementation to bring about the desired cultural change within governmental units (Albukhitan, 2020; Nicolás-Agustín et al., 2022). Hence, a successful implementation of digital transformation could depend on cultural adaptation, and personal skills, as well as overt and covert resistance to the new technology by members of the governmental unit. In line with the technology acceptance model, governmental employees should be aware of the benefits of digital technology. Further, they should improve their ability to use it easily and smoothly. This is considered crucial for an effective adoption of technological innovations (Venkatesh et al., 2003; Venkatesh & Bala, 2008).

Proponents of agency theory assume that agents are inherently self-interested at the expense of principals (owners) (Miller, 2005). The literature argues that digital transformation will overcome the problem of information asymmetry (Benjamin et al., 2024). It facilitates transparency and clarity of the process and leaves behind an audit trail that can be used as a form of an agency monitoring tool (Raza et al., 2023). J. Yang et al. (2024) posited that digital transformation has the potential to enhance information transparency in a way that mitigates opportunistic behavior.

In other words, from the view of the agency theory, the role of digital technology is considered vital for reducing agency costs, limiting managers’ incentives and ability to manipulate accounting information. This issue is vital to protect the interests of investors or to force corporate management to work for the interest of the company, rather than for self-interests, i.e., to reduce agency conflicts in the firm (Danielson & Scott, 2007). This is because digital technology enables all transactions to be tracked and verified easily. Digital technology can also improve the quality of internal control systems, which is vital to improving the quality of accounting information, and, hence, reducing information asymmetry between owners and shareholders (B. Fang et al., 2023; Zhang et al., 2024).

Digital transformation entails a change in the way financial and non-financial information is collected, processed, and disseminated (Shenen, 2023). With digital transformation, it is anticipated that information will be processed faster, operating costs will be lower, and the accuracy and the reliability of accounting systems’ outputs will be enhanced (Marota, 2021; Hadi et al., 2023; Asikpo, 2024; Abhishek et al., 2024b). Thus, digital transformation can contribute to developing the audit profession, for instance, by allowing the audit team to analyze big data using advanced technology with minimal time and physical efforts, generating useful information that enhances the decision-making process (Gepp et al., 2018; F. Fang et al., 2024). In particular, it can enable auditors to verify all transactions, i.e., using a continuous audit approach (Minkkinen et al., 2022). Supporting this view, Fotoh and Lorentzon (2023) emphasized the importance of digital technology in reducing and detecting accounting fraud, decreasing audit costs, and improving audit quality.

Therefore, the concept of digital transformation has recently attracted the attention of many global professional accounting organizations. A study by the Association of Chartered Certified Accountants and the Institute of Management Accountants noted that digital transformation tools can reshape the reality of accounting and auditing professions (ACCA-IMA, 2013). The Institute of Chartered Accountants in England and Wales indicated the value of blockchain technology, as accounting technology, in supporting businesses in general, and the accounting profession in particular (ICAEW, 2018). The American Institute of Certified Public Accountants has also stressed that accountants and auditors should be aware of developments of the blockchain technology and integrate it into more aspects of financial life. Accountants also should be highly prepared to deal with this technology while committing to continuing education (AICPA, 2018). Moreover, the International Federation of Accountants focused on the fundamental concepts of blockchain and how it can simplify processes and change business models, highlighting its potential role for accounting and auditing professionals (IFAC, 2018). In 2020, the International Accounting Standards Board (IASB) identified some areas where digital technologies can make significant changes in the audit profession, such as continuous auditing, using artificial intelligence and machine learning to conduct necessary analyses, and remote auditing (Seidenstein & Davies, 2022).

In developed countries such as the USA and the UK, the positive impact of digital technology on the quality of accounting information systems is well established. In this regard, Morris and Laksmiana (2010) confirmed that earnings management practices decreased among American firms that adopted ERP systems during the period from 1994 to 2003. Other scholars such as Fernandez and Aman (2018) and Kaya et al. (2019) noticed that digital technology sped up data processing, reduced the risk of human errors, and enhanced the efficiency and quality of accounting information systems. Using USA data, S. Yang et al. (2018) observed a significant and consistent improvement in the comparability of financial reports after adopting XBRL. Focusing on Canadian audit firms, Saleh et al. (2023) ensured the importance of big data analysis tools in supporting the quality of financial reports.

The use of digital transformation and its awaited results are highly needed in developing markets with emerging technology-related infrastructures, modest facilities, and weak transparency or low financial reporting quality. Most of the literature has focused on Asian markets, especially China. In this regard, using Chinese data, some scholars have noted the importance of digital transformation in enhancing financial reporting quality (Chen et al., 2016; B. Fang et al., 2023; L. Wang & Hou, 2024). X. Wang et al. (2021) indicated the importance of the Internet of Things technology in reducing real earnings management practices. Z. Wang (2023) demonstrated a positive impact of digital transformation on the transparency of accounting information and on reducing information asymmetry between management and external parties. Mu (2023) indicated a significant positive relationship between digital transformation and the quality of accounting information in Chinese-listed companies. Additional findings indicate that digital transformation enhances the quality of accounting information through two main paths: improving the quality of internal control and attracting the attention of analysts. In the same context, J. Yang et al. (2024) and Zhang et al. (2024) highlighted the value of digital transformation to improve the comparability of accounting information. Also, Papiorek and Hiebl (2024) found that the relationship between the quality of management accounting information systems and management control effectiveness was more pronounced if the firm featured a higher degree of process automation. J. Yang et al. (2024) showed that the impact of digital transformation on improving comparability was more pronounced in non-state-owned enterprises with poor corporate governance quality. Moreover, Asikpo (2024) revealed that digital transformation, including big data analytics, artificial intelligence, blockchain, and cloud computing,

enhanced the quality of financial reporting. [Bin et al. \(2025\)](#) constructed a digital transformation index and examined its impact on information transparency of Chinese enterprises and found that digital transformation greatly enhanced information transparency. Focusing on India, [Abhishek et al. \(2024a\)](#) indicated the positive impact of XBRL technology on enhancing the quality of financial reports.

A few studies have focused on African and Arab markets. For instance, concentrating on Algerian firms, [Oualid et al. \(2021\)](#) examined the requirements of digital transformation, such as the necessary knowledge of the human element, in addition to the effects and obstacles to the digitization of accounting information systems. The results revealed a lack of awareness of the importance of digital transformation and a lack of efforts made to develop accounting information systems in line with the requirements of digital transformation. Focusing on Iraq, [Mikhlef and Smaoui \(2024\)](#) highlighted the significant impact of digital transformation on accounting information systems. Focusing on Jordanian banks, [Al Shanti and Elessa \(2023\)](#) showed that the blockchain technology improved the quality of accounting information and enhances corporate governance systems.

Concerning the implications of new technologies in the public sector, using qualitative data from South Africa, [Chandiwana and Pather \(2016\)](#) showed that the adoption of ERP systems enhanced firms' sustainability, reduced ICT operating costs, resulted in the effective development of the state budget, and led to better governance practices. Using European data, [Bekiaris and Markogiannopoulou \(2023\)](#) found that using ERP systems facilitated the adoption of accrual accounting and enhanced accounting reform efforts in the public sector. [Abubakari and Namooq \(2023\)](#) emphasized the role of digital technology in enhancing accountability and budget implementation and guiding long-term financial planning in the Ghanaian public sector.

Considering the above, it is noted that previous studies in the field of digital transformation and the quality of accounting information have focused more on the private sector. In contrast, studies on the governmental sector are minimal, focusing on the challenges posed by digital transformation, indicating the requirements for administrative and institutional reforms to respond to cyber threats and risks in this important sector ([Eom & Lee, 2022](#)). As a result, some researchers have called for the need to develop governmental financial reports and benefit from the current automation processes to enhance the quality of government financial reports ([Bora et al., 2021](#); [Dos Santos et al., 2024](#); [Kenetey & Popesko, 2024](#)). Thus, we formulated H1 as follows:

H1. *There is a significant positive relationship between digital transformation and the quality of accounting information systems in developing countries' governmental units.*

4. Research Methodology

4.1. Survey Development, Data Collection, and Sample

To study the impact of digital transformation on the quality of accounting information systems in governmental units, we collected data from Saudi Arabia and Egypt. We conducted a survey of the perceptions of financial management managers in governmental entities in the two countries using the survey instrument (as shown in Appendix A). We developed the survey by reference to the academic literature, academics, practitioners, the Saudi Data and Artificial Intelligence Authority, and digital governmental authorities in Egypt and Saudi Arabia. The final electronic version of the survey consisted of open and closed questions about the impact of digital transformation on the quality of accounting information systems in governmental units. We provided respondents with a definition of the concept of digital transformation and its tools so that they could visualize the nature and outcomes of digital transformation in a consistent manner.

Respondents included governmental sector employees from a wide range of sectors in Egyptian and Saudi contexts. Participants were contacted and invited electronically by emailing them a hyperlink to the survey. All participants completed the survey anonymously. The research aimed to seek the opinions of 200 governmental employees in the two countries. However, not all participants responded to the survey. We achieved a participation rate of approximately 64% as we collected 128 complete responses. This rate is consistent with the average rate reported in accounting literature, i.e., 50% (Henri & Thibodeau, 2006). This represented the data included in the regression model about the risks and benefits of digital transformation concerning accounting information systems in various organizations (as shown in Table 1). As explained later in more detail, we examined (a) strategic planning for digital transformation, (b) adequacy of human resources for digital transformation, (c) digital and technological infrastructures, (d) cybersecurity risk management, (e) automated data processing, (f) institutional safeguards for digital transformation, and (g) the quality of accounting information systems in governmental units.

Table 1. Sample distribution.

	Egypt		KSA		Total	
Governmental Units	74 (57.8%)		54 (42.2%)		128	
Sectors	Revenue	Service	Revenue	Service	Revenue	Service
Governmental Units by sector	50 (67.6%)	24 (32.4%)	12 (22.2%)	42 (77.8%)	62 (48.4%)	66 (51.6%)
Characteristics of participants						
Gender	Male		Female		Total	
	100	78.1%	28	21.9%	128	
Experience (years)	<5	5–10	10–15	>15	Total	
	24 (18.8%)	16 (12.50%)	30 (23.40%)	58 (45.30%)	128	
Age (years)	<30	30–39	40–50	>50	Total	
	22 (17.2%)	42 (32.80%)	36 (28.10%)	28 (21.90%)	128	
Qualifications	High school	Bachelor	MSc	Ph.D.	Total	
	8 (6.30%)	46 (35.90%)	42 (32.80%)	32 (25%)	128	

4.2. Variables and Models

4.2.1. Digital Transformation

Creating a digital strategic plan to improve operations is the first step in a successful digital transformation process (Saarikko et al., 2020; Ciancarini et al., 2024). A digital transformation strategy is an organized and formal plan for all activities undertaken by a governmental unit, and it should be based on benefits and risks associated with digital technologies in a comprehensive manner (Albukhitan, 2020). This plan should also include change management requirements for digital transformation to achieve strategic objectives (Jabor & Hamdan, 2023). The executive plan for digital transformation in governmental units must also be developed in accordance with best practices. It must include a detailed list of initiatives and projects related to digital transformation in the governmental unit, the timetable for implementing these initiatives, projects and implementation stages, interim (executive) objectives and performance indicators through which implementation processes are monitored, the extent of progress in achieving strategic objectives of digital transformation, periodic reports and models through which the progress in implementation processes will be measured, and, finally, the mechanism of implementing integration processes in relevant government agencies.

Previous studies have highlighted the value of digital tools in achieving the desired results of digital transformation (Saleh et al., 2023; Aboelfotoh et al., 2024). The relevant infrastructure may include hardware and software, networks, and cloud computing solutions that support the storage, processing, and analysis of large amounts of data (Miaoquan et al., 2023; Ciancarini et al., 2024). Further, databases are essential requirements for implementing digital transformation to achieve easy access to governmental units' data by stakeholders. Here, the standards for data sharing tools and applications that enable citizens to interact with and benefit from data must be effectively developed (Ciancarini et al., 2024).

Governmental units can rely on ERP systems to centralize data collection instead of having scattered databases and organize them so that all users can use central data warehouses securely and manage basic functions, including accounting, project management, risk and compliance management, and supply chain management. This can help in planning, budgeting, forecasting, and reporting on financial achievements, integrating all functions of governmental units, enabling the exchange of data and preventing duplicating data (Jabor & Hamdan, 2023).

In addition, digital transformation requires effective measures to be used by government units to mitigate cyber risks and protect their data, systems, and reputation (Bhimani & Willcocks, 2014). Cybersecurity risk management aims to reduce the risks of malicious attacks on cyberspace and its components of software, computers, and networks, as well as unauthorized access to data and information. This is crucial to ensure confidentiality and integrity, and to detect and thwart intruders. This process can include using tools to detect intrusions and stop viruses, blocking malicious access, enforce authentication, and enable encrypted communications. This, in turn, can protect the cyber environment and assets of the governmental unit and secure the information used by stakeholders. Finally, cybersecurity risk management requires regular reviews of the protection systems in place (Perols, 2019; Miaoquan et al., 2023).

In line with the institutional theory, the management of governmental units may not be fully rational but rather driven by many institutional forces. The governmental unit, as a social and cultural system, is part of the surrounding institutional environment, which puts pressure on its structures and practices and shapes the desired behavior (Schiavi et al., 2024). Institutional forces help corporate management overcome obstacles resulting from workers' resistance to digital transformation within governmental units (Valentinetti & Muñoz, 2021).

Considering the above discussion, we built a *Digital Index* depending on 6 axes established in the questionnaire as follows: strategic planning (SP), human resources management (HRM), digital and technological infrastructure (DTI), cybersecurity risk management (CRM), automated data processing (ADP), and digital transformation institutional guarantees (DTIG). We depended on a 5-point Likert scale, where 1 denotes a very low level, 2 denotes a low level, 3 denotes an intermediate level, 4 denotes an upper intermediate level, and 5 denotes an advanced level.

4.2.2. The Quality of Accounting Information Systems

The use of digital technology is crucial to limit corporate management's ability to manipulate accounting information because all transactions can then be tracked and verified smoothly. This situation can also improve the quality of internal and external control systems, which may ultimately enhance the quality of accounting information (Morris & Laksmama, 2010; X. Wang et al., 2021; Zhang et al., 2024). To examine the dependent variable, the quality of the accounting system in governmental units, we focused on the quality of the accounting system's components and cycles. Adopting a unified information system, as in the case of the governmental resource planning system, requires building

a government accounts' guide electronically as stipulated in the executive regulations of the Unified Public Finance Law issued in 2024, which ensures smooth access to the required information.

We benefited from the standards issued in Egypt and Saudi Arabia in building an index for digital transformation for each government unit based on the evidence document included in each standard of digital transformation, which enables the regulatory authorities to evaluate the performance of governmental units in terms of digital transformation. We examined the impact of digital transformation on the quality of accounting information systems in governmental units using the following proposed model:

$$AISQ_{it} = \alpha + B_1(DigitalIndex_{it}) + B_2(Sector_{it}) + \varepsilon \dots \dots \quad (1)$$

$$AISQ_{it} = \alpha + B_1(DigitalIndex_{it}) + B_2(Sector_{it}) + B_3(Country_{it}) + B_4(country \times DigitalIndex_{it}) + \varepsilon \dots \dots \quad (2)$$

where AISQ is the quality of accounting information systems in governmental agencies, which was measured based on the seven questions included in the questionnaire, as well as the delay in submitting periodic financial reports to the Ministry of Finance. Digital Index is the digital transformation index, which is measured based on the six axes established in the questionnaire (see Appendix A). Sector is a control variable measured as a dummy variable that equals one if the governmental unit works in a revenue sector and zero otherwise. Country is a dummy variable that equals one if the country is Egypt and zero if the country is Saudi Arabia.

5. Empirical Results

5.1. Descriptive Analysis

Table 1 shows the sample distribution. Our sample consisted of 74 (57.8%) governmental units in Egypt, out of which 50 units (67.6%) were in the revenue sector, and 24 units (32.4%) were in the service sector. Also, the sample consisted of 54 (42.2%) governmental units in Saudi Arabia, out of which 12 units (22.2%) were in the revenue sector, and 42 units (77.8%) were in the service sector.

Table 2 reveals the descriptive statistics for the sample. The mean value of AISQ was 2.799 with a standard deviation (SD) of 0.978, and it ranged from 1 to 5. This reflects the lower level of accounting information systems quality in governmental units in the sample. The mean value of the *Digital Index* was 3.183 with an SD of 0.700, and it ranged from 1.32 to 4.43. The mean value of SP was 1.583 with an SD of 0.717, and it ranged from 0.33 to 2.33. The mean value of HRM was 3.335 with an SD of 1.066, and it ranged from 1 to 5. The mean value of DTI was 3.355 with an SD of 1.163, and it ranged from 1 to 5. The mean value of CRM was 3.312 with an SD of 0.611, and it ranged from 2 to 4. The mean value of ADP was 3.445 with an SD of 0.858, it ranged from 1.50 to 4.50. Finally, the mean value of DTIG was 3.390 with an SD of 1.161, and it ranged from 1 to 5. These results reflect the higher variation of digital transformation across governmental units in our sample and indicate that the application of digital transformation is at an intermediate level.

Table 2. Descriptive statistics.

	Mean	Median	SD	Minimum	Maximum
AISQ	2.799	2.857	0.978	1.00	5.00
Digital Index	3.183	3.267	0.700	1.32	4.43

Table 2. *Cont.*

	Mean	Median	SD	Minimum	Maximum
SP	1.583	2.000	0.717	0.33	2.33
HRM	3.335	3.500	1.066	1.00	5.00
DTI	3.355	3.500	1.163	1.00	5.00
CRM	3.312	3.500	0.611	2.00	4.00
ADP	3.445	3.725	0.858	1.50	4.50
DTIG	3.390	3.333	1.161	1.00	5.00

AISQ is the total score computed for the quality of the accounting information system in government units' questions. *Digital Index* is the total score computed for the six categories of questionnaires related to digital transformation. *SP* is the total score for strategic planning and follow-up questions. *HRM* is the total score for human resources management questions. *DTI* is the total score for digital and technological infrastructure questions. *CRM* is the total score for cybersecurity risk management questions. *ADP* is the total score for automated data processing questions. *DTIG* is the total score for digital transformation institutional guarantees questions.

Table 3 shows Cronbach's alpha test, where the alpha score was 0.926 for questions that measured *Digital Index* in governmental units. This shows that the reliability of these questions was high (square root of the alpha was 96.23%). The alpha score was 0.884 for questions that measured *AISQ* in governmental units, reflecting the reliability of these questions (square root of the alpha was 94.02%). The overall alpha score for all questions was 0.836, reflecting the reliability of these questions (square root of the alpha was 91.43%).

Table 3. Cronbach's alpha test.

	Cronbach's Alpha		
	Egypt	KSA	Total
Digital Index Questions	0.917	0.924	0.926
AISQ Questions	0.881	0.885	0.884
All sample Questions	0.842	0.797	0.836

Before employing the factors in additional analyses like multiple regression or multi-variate analysis of variance, Factor Analysis is useful for reducing the number of factors to be more manageable. This is considered useful when creating a questionnaire. Including more statements in the questionnaire does not provide a thorough grasp of the factors. Unrelated items can be eliminated from the final questionnaire with the aid of Factor Analysis. Table 4 shows a higher efficiency of all questionnaire statements (Component 1 is greater than 0.40 for all the statements) in measuring what they were designed to measure.

Table 5 shows the preliminary statistical variance between *AISQ* and *Digital Index* in Egypt and Saudi Arabia. The reported findings show that *AISQ* was higher in Egypt. However, these differences were insignificant. The results indicate that the *Digital Index* in Saudi Arabia was significantly higher than Egypt at the 1% significance level. This may be because Saudi Arabia is one of the richest countries in the MENA region and in the world, and it has massive financial resources that can support the digital transformation process in the public sector.

Table 4. Factor Analysis.

Factor Analysis for Dependent and Independent Variables	Component	Factor Analysis for All Digital Index Dimensions		Component
	1			1
Digital Index Questions				
Q1	0.730	SP	Q1	0.984
Q2	0.725		Q2	0.984
Q3	0.812		Q3	0.924
Q4	0.771	HRM	Q4	0.750
Q5	0.795		Q5	0.901
Q6	0.743		Q6	0.903
Q7	0.706		Q7	0.875
Q8	0.744	DTI	Q8	0.819
Q9	0.781		Q9	0.872
Q10	0.782		Q10	0.925
Q11	0.821		Q11	0.893
Q12	0.931	CRM	Q12	0.852
Q13	0.851		Q13	0.739
Q14	0.881		Q14	0.750
Q15	0.881		Q15	0.750
Q16	0.852	ADP	Q16	0.955
Q17	0.882		Q17	0.954
Q18	0.838		Q18	0.909
Q19	0.832		Q19	0.912
Q20	0.739	DTIG	Q20	0.860
Q21	0.795		Q21	0.912
Q22	0.861		Q22	0.939
AISQ Questions				
Q23	0.867			
Q24	0.854			
Q25	0.840			
Q26	0.814			
Q27	0.798			
Q28	0.775			
Q29	0.490			

Table 5. Independent sample *t*-test.

Variables	Egypt		KSA		<i>t</i> -Test	Sig
	Mean	SD	Mean	SD		
AISQ	2.876	0.948	2.693	1.017	1.048	0.297
Digital Index	3.011	0.664	3.420	0.683	−3.399 ***	0.001
SP	1.468	0.726	1.741	0.681	−2.151 **	0.033
HRM	3.061	1.049	3.713	0.980	−3.570 ***	0.001
DTI	3.068	1.127	3.750	1.105	−3.411 ***	0.001
CRM	3.203	0.656	3.463	0.513	−2.423 **	0.017
ADP	3.285	0.896	3.665	0.758	−2.523 **	0.013
DTIG	2.982	1.089	3.951	1.023	−5.097 ***	0.000

, * are significant at the 5, and 1 percent levels, respectively.

5.2. Regression Analysis

Table 6 shows the linear regression results. The results indicate that *Digital Index* had a negative effect on *AISQ* when this effect was examined for the whole sample (Egypt and Saudi Arabia) (coefficients = −0.687, −0.509, −1.018 for all sample, Egypt and Saudi Arabia, respectively, $p < 0.01$). This finding indicates the existence of a negative effect of digital transformation on the quality of accounting information systems in the two countries' public sectors, and hence, our hypothesis is rejected. This finding can be understood concerning the immaturity of using digital technologies in the two countries' public sectors, which is yet to bring positive implications for accounting and auditing professions. This result is consistent with [F. Fang et al. \(2024\)](#), who noted that during initial stages of digital transformation, i.e., when the transformation level is relatively low, the complexity of digital systems may negatively affect the accounting information systems and audit quality. In addition, Model 2 shows that the negative effect of digital transformation was lower in Egypt, as *Digital Index* × *Country* had a positive effect on *AISQ*. This finding reflects that the negative effect of the *Digital Index* on *AISQ* was mitigated in Egypt. It also indicates that Egypt had personnel more qualified in using and benefiting from digital technologies compared to the situation in Saudi Arabia.

Table 6. Linear regression results.

Variables	Model (1)		Model (1)		Model (1)		Model (2)	
	All Sample		Egypt		KSA		All Sample	
	B	t	B	t	B	t	B	t
Intercept	4.976	13.813 ***	4.272	8.625 ***	6.184	11.434 ***	6.266	10.540 ***
Digital Index	−0.687	−6.316 ***	−0.509	−3.279 ***	−1.018	6.362 ***	−1.054	−6.110 ***
Sector	0.024	0.159	0.210	0.945	−0.031	−0.119 *	0.141	0.826
Country							−0.212	−1.174
Digital Index * Country							0.579	2.562 **
F. Test	19.949 ***		5.667		23.066		12.224 ***	
Adjusted R ²	23%		11.3%		45.4%		26.1%	
Observations	128		74		54		128	
VIF	Less than 4		Less than 4		Less than 4		Less than 4	

*, **, *** are significant at the 10, 5, and 1 percent levels, respectively.

Table 8. Cont.

Variables	Model (2)		Model (2)		Model (2)		Model (2)		Model (2)		Model (2)		Model (2)	
	B	t	B	t	B	t	B	t	B	t	B	t	B	t
SP × Country			0.359	1.672 [*]										
HRM × Country					0.362	2.342 ^{**}								
DTI × Country							0.248	1.692 [*]						
CRM × Country									0.877	3.166 ^{***}				
ADP × Country											0.620	3.489 ^{***}		
DTIG × Country													0.440	3.404 ^{***}
Gender	−0.426	−2.340 ^{**}	−0.377	−1.859 [*]	−0.422	−2.083 ^{**}	−0.337	−1.628	−0.297	−1.424	−0.340	−1.724 [*]	−0.414	−2.134 ^{**}
Age	0.083	0.691	0.066	0.493	0.025	0.180	0.101	0.747	0.109	0.810	0.083	0.640	0.081	0.634
Experience	−0.049	−0.356	−0.032	−0.214	−0.028	−0.187	−0.010	−0.063	−0.047	−0.308	−0.017	−0.114	−0.008	−0.053
Qualifications	0.174	1.717 [*]	0.202	1.733 [*]	0.137	1.208	0.148	1.290	0.082	0.707	0.099	0.892	0.127	1.176
F. Test	7.591 ^{***}		2.998 ^{***}		3.062 ^{***}		2.776 ^{***}		3.042 ^{***}		4.249 ^{***}		4.932 ^{***}	
Adjusted R ²	29.30%		11.20%		11.50%		10.10%		11.40%		17%		19.85%	
Observations	128		128		128		128		128		128		128	
VIF	Less than 4		Less than 4		Less than 4		Less than 4		Less than 4		Less than 4		Less than 4	

^{*}, ^{**}, ^{***} are significant at the 10, 5, and 1 percent levels, respectively.

6. Discussion

This study explored how digital transformation affects accounting and auditing professions and the quality of the accounting information system in governmental units in two developing contexts in the MENA region—Egypt and Saudi Arabia. As explained in Section 5.2, we found that *Digital Index* had a negative effect on *AISQ* when this effect was examined for the whole sample (Egypt and Saudi Arabia) (coefficients = −0.687, −0.509, −1.018 for all sample, Egypt and Saudi Arabia, respectively, $p < 0.01$). Thus, in contrast to many studies in the literature, especially those conducted in the USA and Chinese contexts (e.g., Morris & Laksmana, 2010; S. Yang et al., 2018; Chen et al., 2016; X. Wang et al., 2021; B. Fang et al., 2023; L. Wang & Hou, 2024), our analyses revealed a negative effect of digital transformation on the quality of accounting information systems in the two countries, which could be interpreted concerning the premature situation of digital technologies in emerging markets.

Further, the negative effect of digital transformation was lower in Egypt, as the *Digital Index* × *Country* had a positive effect on *AISQ*. This finding reflects that the negative effect of the *Digital Index* on *AISQ* was mitigated in Egypt. In other words, the negative effect of digital transformation was found to be lower in Egypt compared to Saudi Arabia. The current findings indicate that there are likely to be negative effects on the quality of the accounting information system in the early stages of digital transformation, i.e., when the organization's digital technology is in the growth stage, and the information subsystems across different departments are complex and unconnected, as is the case in many emerging markets, including Egypt and Saudi Arabia. Such complexity allows management to engage in opportunistic practices that may harm the quality of accounting information systems (F. Fang et al., 2024; Abhishek et al., 2024b).

Our findings reveal that during early stages of digital transformation, which is the case in many emerging markets, including Egypt and Saudi Arabia, information across different governmental departments is not highly interconnected. The existing complexity

is likely to create technical barriers for auditors to fully understand digital issues in the governmental unit and access appropriate information to verify the truthfulness and fairness of data and information, especially when auditors are not highly qualified to review existing digital technology (F. Fang et al., 2024). In contrast, when the digital transformation is at an advanced stage, and the information infrastructure is complete, there will be an integration of information systems in different departments. This integration provides the supervisory authorities with the ability to analyze big data, which enhances the timeliness, and effectiveness of obtaining the required information. Consequently, it is inferred that the transparency of information in governmental units and the supervisory efficiency of the authorities become more effective during advanced stages of digital transformation (F. Fang et al., 2024). Hence, consistent with the contingency theory, we believe that the impact of digital transformation on the quality of accounting and auditing information in governmental units is conditional on the level or the stage of digital transformation. Moreover, from the agency theory perspective, the positive impacts of the digital transformation concerning the quality of accounting information is considered vital to help investors depend on reliable information in their investment decisions and, hence, minimize the existence of information asymmetry between owners and managers (Sun et al., 2022).

7. Conclusions

This study addressed the impact of digital transformation on the quality of the accounting information system in governmental units in two developing countries—Egypt and Saudi Arabia. The current findings contribute to the literature by focusing on the governmental sector and investigating a new context—the MENA region—rarely examined in previous studies. Moreover, our findings can have implications for regulators in the public sector for developing governmental financial reports by benefiting from the digital transformation process. Our findings imply that regulatory bodies in emerging markets should keep pace with recent developments in the digital environment. They should manage the challenges brought about by digital transformation in accounting and auditing professions. Moreover, it has become necessary for public sectors' accountants and auditors in Egypt and Saudi Arabia to adapt to and benefit from the new digital environment, and gain the required proficiency in dealing with, governing, and reviewing the new technological tools.

However, our study's findings are not without limitations which may limit their generalizability and, hence, these limitations should be considered while interpreting such findings. These limitations open avenues for future research, as follows. Considering the focus of this study on two particular MENA countries, it is suggested that future research can examine other developing contexts to ensure the generalizability and the relevance of the current findings for other global contexts beyond the MENA region. Further, regarding the limitations of solely depending on surveys in collecting data, we suggest that future research can use other data collection and research methods, especially interpretative methods, to reveal the rationales behind the current findings. Further, future research could depend on longitudinal analysis to add further depth and transparency regarding the long-term effects of digital transformation as systems mature. Finally, it is suggested that future research can draw upon other measures of digital transformation and accounting information systems' quality for a fuller understanding of the link between them.

Author Contributions: Conceptualization, A.M.E. and A.H.; methodology, A.H.; writing—original draft, A.H. and A.M.E.; writing—review and editing, A.D.; project administration, A.H. and A.D. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki, and the protocol was approved by the Ethics Committee of Prince Sultan University (Project identification code: PSU IRB-2025-02-0218).

Informed Consent Statement: Informed consent for participation was obtained from all subjects involved in the study.

Data Availability Statement: The data presented in this study is available on request from the corresponding author.

Acknowledgments: The authors would like to thank Prince Sultan University for their support.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

Table A1. The survey.

(1) General Data			
-Country:			
Egypt		Saudi Arabia	
-Sector:			
Service (Education- Health- Justice- Local Administration etc.)		Revenue (Ministry of Finance- Real Estate Registry-Government Banks- Ministry of Petroleum-Electricity-Water and Utilities.etc.)	
-Gender:			
Male		Female	
-Experience:			
Less than 5 years	From 5 to 10 years	From 10 to 15 years	More than 15 years
-Age:			
Less than 30 years	From 30 to 39 years	From 40 to 49 years	50 years and above
-Educational level:			
High school or equivalent	Bachelor's/Licentiate	Master's	Doctorate
(2) Questions			
1-Digital Transformation Strategic Planning and Follow up			
1. Do you have strategic planning for digital transformation? (yes or no)			
2. Do you have operating plans for digital transformation? (yes or no)			
3. To what extent does the government unit follow up on the strategic and operating plans for digital transformation? (5-point Likert scale)			
2-Human Resources Management			
4. To what extent are the material and human resources available to the unit sufficient to implement digital transformation? (5-point Likert scale)			
5. To what extent are the unit's employees prepared, trained and developed to ensure their acceptance and support of digital transformation? (5-point Likert scale)			
6. To what extent are the advantages and benefits of digital transformation explained to the unit's employees by managers and heads? (5-point Likert scale)			
7. To what extent are there organized training processes for the unit's employees on digital transformation processes? (5-point Likert scale)			

Table A1. Cont.

(2) Questions
3-Digital and technological infrastructure
8. To what extent does the current digital and technological infrastructure help the unit in enhancing the concept of digital transformation? (5-point Likert scale)
9. To what extent does the government unit rely on platforms to provide services to the public and stakeholders? (5-point Likert scale)
10. To what extent does the government unit have integrated databases? (5-point Likert scale)
11. To what extent does the government unit apply advanced technology tools that support digital transformation such as: government resource planning systems–blockchain technology.etc.? (5-point Likert scale)
4-Cybersecurity Risk Management
12. To what extent does the government unit rely on implementing effective procedures to manage database cybersecurity risks? (5-point Likert scale)
13. To what extent does the government unit face database cybersecurity breaches and risks? (Reversed) (5-point Likert scale)
14. In the event of previous periods having cybersecurity breaches and risks, how significant were those breaches? (Reversed) (5-point Likert scale)
15. To what extent does the unit work to limit inappropriate access to information and secure data and information? (5-point Likert scale)
5-Automated data processing
16. To what extent are customer transactions entered directly into the system? (5-point Likert scale)
17. To what extent are employee transactions entered directly into the system? (5-point Likert scale)
18. To what extent are supplier transactions directly entered into the system? (5-point Likert scale)
19. To what extent is all data automated? (5-point Likert scale)
6-Digital transformation institutional guarantees
20. To what extent are digital transformation processes in the government unit monitored by the Ministry of Finance, the Ministry of Communications and other relevant authorities? (5-point Likert scale)
21. To what extent do you see that the current legislative and legal amendments enhance the success of digital transformation in the government unit? (5-point Likert scale)
22. To what extent is the government unit committed to the governance of digital transformation processes? (5-point Likert scale)
7-Quality of the accounting information system in government units
23. To what extent are the financial irregularities and accounting errors discovered by the internal auditor during the year significant? (Reversed) (5-point Likert scale)
24. To what extent are the financial irregularities and accounting errors discovered by the external auditor during are significant? (Reversed) (5-point Likert scale)
25. To what extent are the financial irregularities and accounting errors discovered by the oversight bodies during the year significant? (Reversed) (5-point Likert scale)
26. To what extent are the corruption operations reported during the year by employees significant? (Reversed) (5-point Likert scale)
27. To what extent were the internal cybersecurity breaches and threats that occurred during the year significant? (Reversed) (5-point Likert scale)
28. To what extent were the external cybersecurity breaches and threats that occurred during the year are significant? (Reversed) (5-point Likert scale)
29. To what extent is there a delay in submitting periodic reports to the Ministry of Finance? (Reversed) (5-point Likert scale)

Notes

- ¹ See <https://tahseen.ae/insights/challenges-and-opportunities-for-egypt-s-evolving-digital-economy/> (accessed on 1 February 2025) for more details.
- ² For more details, please refer to the Ministry of Communications website, Communications Sector Strategy https://mcit.gov.eg/ar/Digital_Egypt (accessed on 1 February 2025).
- ³ See https://mcit.gov.eg/ar/Digital_Egypt (accessed on 1 February 2025) for detail.
- ⁴ For more details, see the Basic Standards for Digital Transformation (2024), Digital Government Authority, Kingdom of Saudi Arabia (<https://www.my.gov.sa/wps/portal/snp/aboutksa/digitaltransformation/?lang=en>), accessed on 1 February 2025).

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